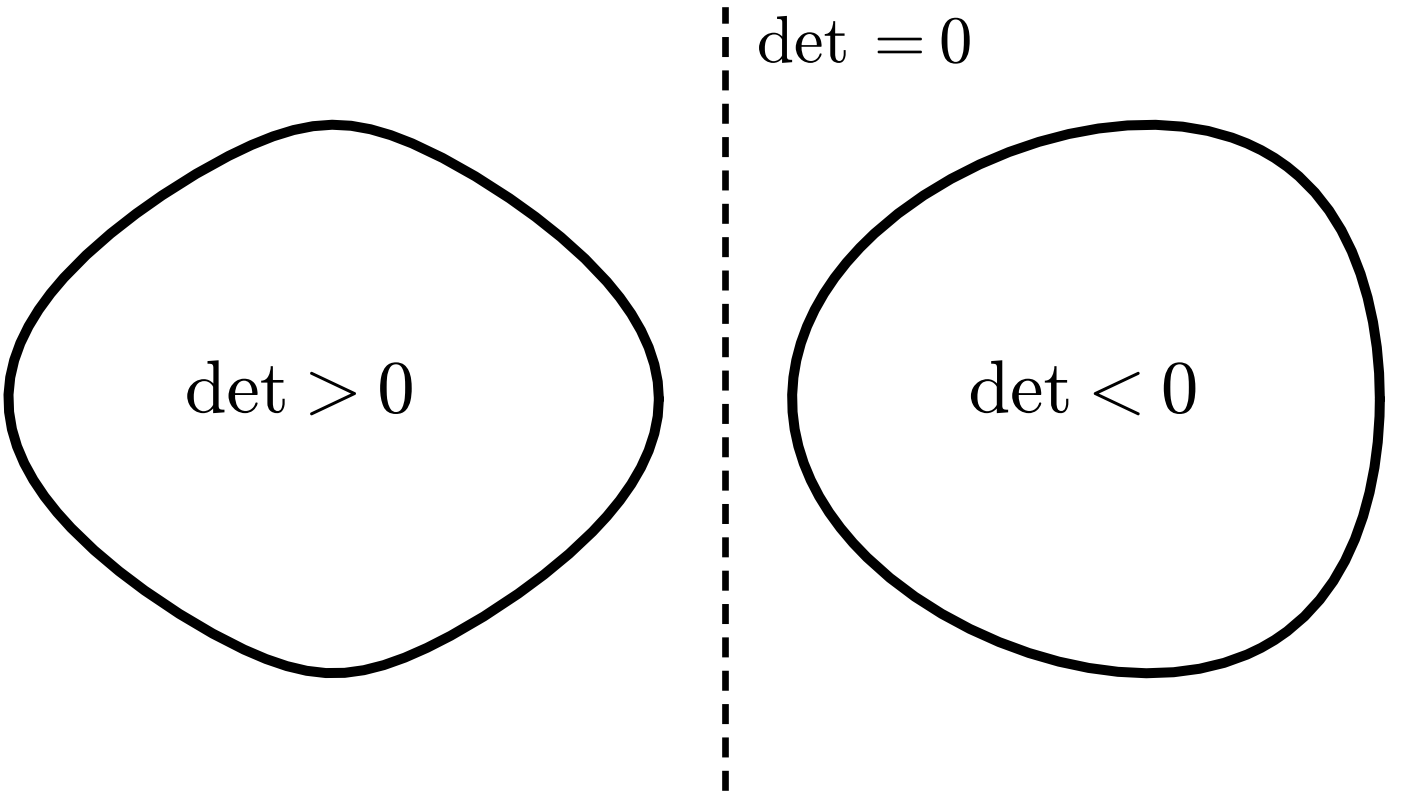
$\det = 0$
$\det > 0$
$\det < 0$